



# When researchers use AI: public trust, ethical judgments, and the perceived value of academic research

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## Abstract

As generative artificial intelligence (AI) tools become increasingly integrated into scientific research, questions arise about how such integration affects perceptions of legitimacy, accountability, and fairness in the production of scientific knowledge. This study investigates how the disclosure of AI use in academic research shapes public perceptions of researchers and their work. In a preregistered survey experiment ( $N=806$ ), respondents were randomly assigned to read one of six author disclosure statements describing different levels and types of generative AI use. Respondents reported the highest levels of trust, perceived research quality, and perceived ethicality when the research was described as conducted entirely without AI, and the lowest when AI was used for both theoretical and methodological tasks. Respondents were more positive toward the use of generative AI when it was used for methodological tasks such as coding compared with theoretical tasks such as hypothesis generation. “AI for language polishing” was perceived as least transparent. Frequency of AI use, but not self-reported AI knowledge or general trust in science, moderated these effects: individuals with more personal experience using AI were more accepting of AI-assisted research. These results indicated that public aversion to AI extended into the scientific domain and is driven less by ignorance than by concerns over diminished human creativity, effort, and accountability. Implications for maintaining trust and engagement in science amid increasing AI integration are discussed.

**Keywords** AI aversion · Transparency · Disclosure · Public trust in science · Science communication

## 1 Introduction

Recent advances in generative AI have transformed how people work, think, and create. From business and law to medicine, education, and creative industries, generative AI systems are increasingly adopted to enhance productivity, streamline repetitive tasks, and augment human decision-making [4, 17, 45]. The academic world is no exception. Researchers are beginning to use generative AI to accelerate

literature re-views, brainstorm research ideas, design experiments, and polish manuscripts, potentially making scientific production faster and more inclusive [3, 32, 53]. Recent reviews and commentaries highlight the potential for LLMs to speed discovery and reduce experimental costs while also cautioning about limits to reliability and reproducibility when models are scaled without careful guardrails [62, 63].

However, growing evidence suggests that human reactions to AI-assisted work are often ambivalent. A growing body of research centers on AI aversion, the tendency for humans to reject outputs or products when they know AI was involved [44]. AI aversion has been documented across a wide range of domains, including medical diagnosis [33], human resources decisions [38], business decisions [5], or news production [34]. AI aversion is expressed even when AI systems perform as accurately as or even better than humans.

As generative AI enters the domain of scientific research, similar concerns arise: when AI assists in the creation of knowledge, does it alter how that knowledge—and its

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human authors—are perceived? Should the involvement of AI be disclosed, and if so, what are the consequences for perceptions of scientific credibility, authorship, and value? These questions have become especially salient amid a broader climate of declining public trust in science and scientists [9, 21, 22]. Since trust underlies the credibility of scientific communication, any factor that alters perceptions of who produces knowledge, such as the use of AI in academic writing, may influence whether audiences find research findings or researchers believable, competent, or valuable.

To address these issues, the current study provides a pre-registered systematic experimental test ( $N=806$ ) of how different types of AI disclosure affect a broad array of credibility and quality judgments about academic research. We assessed (a) trust, both trust in the researcher and trust in the findings; (b) competence and quality—perceived researcher skillfulness and perceived quality of the paper; (c) level of engagement—interest, importance, usefulness, and willingness to share; and (d) perceived ethicality and transparency of research.

### 1.1 AI involvement, perceived credibility, and audience engagement

As artificial intelligence tools become increasingly integrated into professional and academic settings, scholars have begun to examine how audiences perceive outputs created with or by AI and people who adopted AI in this process [8, 11, 44]. A key factor shaping these perceptions is disclosure, that is, whether people know that AI was involved. Disclosure serves as a necessary premise for comparison: only when audience know that AI contributes to a product or process can they meaningfully evaluate it relative to human-produced work.

Research on AI-generated products generally finds that disclosing AI involvement reduces perceived credibility and quality compared to human-only counterparts. For example, in communication and journalism contexts, content on socio-political topics disclosed as AI-authored is perceived as less credible [61]. Likewise, science journalism articles labeled as AI-written significantly decreased perceived credibility of the work [29]. However, the relationship between AI disclosure and evaluation is not uniformly negative. Some studies have found minimal or null effect of AI-generated content disclosure on message credibility or accuracy. For example, disclosing AI involvement did not reduce perceived article quality [61], and labeling content as AI-generated did not significantly change perceived accuracy, credibility, or sharing intentions in general [30].

Additionally, studies examining people who use AI show that disclosure of AI assistance can negatively influence judgments of the human creator. Individuals who reveal

using AI are often perceived as less competent, motivated, or trustworthy than those who do not [8, 11, 44, 54]. These social evaluations can extend beyond specific tasks to broader assessments such as hiring or promotion decisions, suggesting that AI use may carry reputational costs even when performance remains unchanged [47].

The negative consequences of AI disclosure parallel findings from other transparency contexts. For instance, research on conflict-of-interest disclosures has shown that while such transparency is intended to promote ethical accountability, it can paradoxically reduce trust in the discloser [50, 51]. Similarly, revealing that one's work involved AI may signal a lack of independence, authenticity, or human expertise, thus diminishing perceived credibility and quality.

In addition to influencing credibility judgments, disclosure of AI involvement can shape audience engagement: people's interest, attention, and willingness to interact with or share AI-assisted content. For example, when a doctor disclosed using AI for diagnosis, patients reduced intentions to seek help [8]. Similarly, disclosure of AI use in charity ads decreased donation intention [2]. News headlines labeled as AI-generated were shared less often, even when the headlines were actually human-generated [1]. Such lower engagement can be explained by reduced trust [8], diminished empathy [2], or perceived lower human intellectual input [1].

Building on this literature, it is reasonable to expect that when evaluating scientific work, the public will perceive research conducted solely by human authors as more credible, trustworthy, and skillful than research assisted or generated by AI tools. If prior findings on trust and engagement generalize to scientific communication, disclosing AI involvement may signal reduced human expertise, authenticity, and accountability, thereby diminishing perceived research quality and audience interest and engagement. Hence, we hypothesize:

**H1** *People will report the highest levels of trust, perceived competence and quality, and level of engagement towards researchers and research when they learn that the research was conducted entirely by the authors without any help from any generative AI tools.*

### 1.2 Not all AI use is equal: varying roles and perceptions

Philosophers of science have long held that the generation of novel theories and bold hypotheses is a defining feature of scientific creativity and progress. As Popper [43] argued, scientific advancement is viewed as a process of conjectures and refutations, which places hypothesis generation at the epistemic core of scientific inquiry. Conceptual and

theoretical breakthroughs are seen as the foundation of scientific originality and drive paradigm shifts [27, 56] that reshape the fields of knowledge. This perspective is not confined to philosophers of science alone: computational and cognitive approaches have likewise modeled hypothesis generation as a reasoning process, demonstrating that creative theory construction can be systematized and empirically examined. Such work further supported the view that theoretical reasoning is a central component of scientific originality [58].

Scientists' major contributions typically involve new conceptual combinations, theoretical reconceptualizations, or novel hypotheses that open up fruitful research directions [57, 59]. From this perspective, the question is not only whether researchers' use of generative AI matters, but also how and to what extent they integrate it into the research process. The perceived depth of AI involvement in research may shape audiences' judgments about the credibility, quality, and authorship of scientific work, and the evaluation of authors. The degree of AI's contribution, whether reflected in varying levels of assistance (e.g., low, medium, or high) or in the extent of human authors' control over the generated content, significantly influences perceptions of human ownership and the perceived quality of the work [16, 20, 31]. Accordingly, the use of generative AI in theoretical or conceptual tasks may be perceived as a greater threat to authors' intellectual ownership and quality than its use in methodological roles and language roles. Hence, we make the following hypotheses:

**H2** *People will report lower level of trust, perceived competence and quality, and level of engagement towards researchers and research when they learned that researchers used Gen AI for theoretical task (framing, reviewing literature, and generating hypotheses) compared with methodological task (statistical analyses and coding).*

As a logical extension:

**H3** *People will report the lowest levels of trust, perceived competence and quality, and level of engagement towards researchers and research when researchers use Gen AI for both theoretical tasks and methodological tasks.*

### 1.3 AI for language polishing

A growing body of research is positive toward the use of generative AI to polish language. AI-powered writing tools can help reduce language barriers in academic writing, especially for researchers whose first language is not English [26]. Given the dominance of English in academic publishing, language-polishing generative AI tools can

help reduce linguistic disadvantages between native and non-native speakers and make scientific knowledge more broadly accessible [28]. Masters [37] even explicitly said, "researchers whose mother tongue is not English now have a fairer footing" (p.1564).

Nonetheless, the extent of language polishing matters. Altay and Gilardi [1] found that if participants were told AI selected the topic and wrote the whole article, engagement with the article significantly dropped. When participants were told that AI was used only to improve clarity or style, labeling content as AI-generated did not reduce perceived accuracy or willingness to share.

In academic settings, people cherish the human intellectual input. It is reasonable to posit that explicitly clarifying that critical intellectual components of the research were conducted solely by the authors reinforces perceptions of human intellectual contribution and credibility. Such clarification signals that the authors still invested their own expertise and effort and remain accountable for the scientific claims. AI was used only as a writing assistant, similar to a spellchecker or copy editor. If readers accept this signal, they should view the research as more competent, higher quality, and more valuable, and they may feel more trust and engagement with both the researchers and their work.

**H4a** *Compared with research using Gen AI for language polishing ("The authors used generative AI tools to assist with language polishing."), adding "The theoretical framing, hypotheses, analyses, and interpretation were developed entirely by the authors without any help from any generative AI tools." increases people's levels of trust, perceived competence and quality, and level of engagement towards researchers and research.*

On the other hand, the AI aversion literature suggests a more cynical possibility. People often distrust AI users, perceiving them as less competent or less qualified [8, 11, 47]. Once AI use is disclosed, readers may already hold a general concern that "any AI involvement is risky" and form a negative impression of the AI users. In this context, adding the statement that "The theoretical framing, hypotheses, analyses, and interpretation were developed entirely by the authors without any help from any generative AI tools." can draw even more attention to AI use, further heightening the salience of AI use rather than alleviating concern. Readers could think, "If researchers need AI to polish language, maybe they are not very skilled writers or careful communicators." This resonates with recent research indicating systemic biases against "AI-like" writing style: academic reviewers believed "AI-like" writing style signaled non-native authorship and then associated the use of AI tools with lower scientific quality and trustworthiness [28].

The extra emphasis that AI did not help with intellectual tasks might itself trigger suspicion. Rather than reassuring readers, the clarification could be interpreted as defensive, prompting readers to speculate that AI influenced the research more than acknowledged and that the authors are attempting to downplay its role. Together, these reactions suggest that additional clarification about limited AI use may backfire by amplifying doubts about researcher competence and transparency.

**H4b** *Adding “The theoretical framing, hypotheses, analyses, and interpretation were developed entirely by the authors without any help from any generative AI tools.” may lower people’s levels of trust, perceived competence and quality, and level of engagement towards researchers and research. H4a and H4b are competing hypotheses.*

## 1.4 The roles of AI experience and knowledge

Empirical evidence suggests that AI experience, often operationalized as the frequency of use, tends to reduce skepticism toward AI-assisted work. For example, Chen and Cui [8] found that the so-called “AI trust gap”, the differential trust placed in human-only versus AI-assisted work, diminishes as individuals’ AI use increases. Their moderation analysis showed a positive interaction between AI use frequency and trust, indicating that more frequent users are better able to calibrate their trust and may become more comfortable with AI involvement in tasks. This aligns with the broader literature on learned trust in automation, where repeated, successful interactions lead to more calibrated and often higher trust [6, 24, 25].

In contrast, AI knowledge—particularly self-reported knowledge or literacy about AI—can have a more complicated and sometimes counterintuitive effect. Some studies demonstrate that greater AI knowledge is associated with increased skepticism or distrust toward AI systems and AI-assisted work. For instance, Pan et al. [41] observed that participants with higher knowledge of generative AI were more inclined to distrust AI as a collaborator, possibly reflecting heightened awareness of AI’s limitations or ethical concerns [25].

Further complicating this picture, Said et al. [52] found that while AI knowledge influences perceptions of risks and benefits, people with more knowledge exhibited a form of “risk blindness,” underestimating potential risks associated with AI, and at the same time form a more positive perception of AI. Similarly, Lucas et al. [35] examined trust in AI among K-12 teachers and found that AI knowledge significantly predicted trust in AI systems.

Given the more consistent evidence linking experience to reduced distrust and greater calibration of evaluations,

we pre-registered AI use frequency as the primary moderator of public perception of AI-assisted research. The role of AI knowledge was included only as an exploratory factor. Hence, we hypothesize:

**H5** *The trust gap will be smaller among people who use Gen AI more often. People with more experience in Gen AI will be more lenient on research and researchers that use Gen AI.*

## 2 Method

### 2.1 Participant

Participants ( $N=806$ ) were recruited through Prolific, an online platform widely used for survey and experimental research. Prolific was selected because prior studies have shown that its participants tend to demonstrate higher attentiveness, provide more thoughtful responses, follow instructions more closely, and recall previously presented information accurately [14, 40]. Prolific employed quota sampling such that the sample reflected the demographic composition of the U.S. population with respect to gender, age, and ethnicity. A comparison between the demographics of the study participants and those of the U.S. population is available in the [Supplementary Material](#).

### 2.2 Sample size rationale

In Chen and Cui [8], the partial  $R^2$  of the interaction term of treatment status and frequency of AI use was 0.01. We expect a similar effect size for the current research. An a priori power using G-Power 3.1 [18] suggests that a sample size of 780 has a 80% power to capture partial  $R^2$  of 0.01 with  $\alpha=0.05$ . All tests are two-tailed.

The interaction between treatment status and frequency of AI use was the smallest focal effect that we would predict. The main effects in  $H_1$ – $H_4$  were expected to be larger and therefore require a smaller sample. Accordingly, we selected a target sample size of 780.

### 2.3 Experiment design

Participants were randomly assigned to one of six conditions. Respondents saw, “You are reading a paper published in an academic journal. Below is the authors’ statement about whether or how they used generative AI in writing the paper,” followed by a statement from the authors about whether and how generative AI was used in the paper. Participants were not shown an actual research article. The six conditions are as follows:

1. *No AI use* ( $N=126$ ): “This research was conducted entirely by the authors without any help from any generative AI tools.”
2. *Language polishing* ( $N=128$ ): “The authors used generative AI tools to assist with language polishing.”
3. *Language polishing with additional clarification* ( $N=150$ ): “The authors used generative AI tools to assist with language polishing. The theoretical framing, hypotheses, analyses, and interpretation were developed entirely by the authors without any help from any generative AI tools.”
4. *Theoretical assistance* ( $N=120$ ): “The authors used generative AI tools to help with theoretical tasks such as framing, reviewing literature, and generating hypotheses. All methods, statistical analyses, and coding were developed by the authors.”
5. *Methodological assistance* ( $N=148$ ): “The authors used generative AI tools to help with methodological tasks such as statistical analyses and coding. All theoretical framing, literature review, and hypotheses were developed by the authors.”
6. *Both theoretical and methodological assistance* ( $N=134$ ): “The authors used generative AI tools to help with both theoretical tasks (framing, literature review, hypothesis generation) and methodological tasks (statistical analyses and coding).”

After reading the statement, participants reported perceived trust, competence, quality, interest, importance, usefulness, ethicality, transparency, and likelihood of sharing. Respondents also reported their trust in science, knowledge about generative AI, frequency of using generative AI, and demographics.

This research was conducted under the approval of Louisiana State University Institutional Review Board (IRBAM-25-0997). Informed consent was obtained from all respondents. No personally identifiable information was collected. Participant responses were stored and analyzed in deidentified format. Participants received compensation consistent with Prolific’s fair pay guidelines. Each participant was compensated at or above the equivalent of \$10.59 per hour, consistent with fair wage standards.

## 2.4 Measures

### 2.4.1 Dependent variables

All dependent variables and moderators were coded as a numeric variable ranging from 0 to 1 with equal intervals in between.

**2.4.1.1 Trust** *Trust in researchers*: Measured with “To what extent do you trust the authors of the paper?” The response options were “None at all,” “A little,” “A moderate amount,” “A lot,” and “A great deal.”

*Trust in findings*: Measured with “To what extent do you trust the findings of the paper?” The response options were “None at all,” “A little,” “A moderate amount,” “A lot,” and “A great deal.”

**2.4.1.2 Competence/quality** *Perceived skillfulness of researcher*: Measured with “How skilled do you think the authors are at conducting research?” The response options were “Not skilled at all,” “Slightly skilled,” “Moderately skilled,” “Very skilled,” and “Extremely skilled.”

*Perceived quality of paper*: Measured with “How high or low quality do you consider this paper overall?” The response options are “Extremely low quality,” “Moderately low quality,” “Somewhat low quality,” “Neither high nor low,” “Somewhat high quality,” “Moderately high quality,” and “Extremely high quality.”

**2.4.1.3 Level of engagement** *Level of interest in research*: Measured with “How interested are you in reading this research?” The response options were “Not at all interested,” “Slightly interested,” “Moderately interested,” “Very interested,” and “Extremely interested.”

*Perceived importance of research*: Measured with a five-point Likert scale question: “How important do you think this research is?” The response options are “Not at all important,” “Slightly important,” “Moderately important,” “Very important,” and “Extremely important,” and will be coded as 0, 0.25, 0.5, 0.75, and 1, respectively.

*Perceived usefulness of research*: Measured with “How useful do you think this research is?” The response options were “Not at all useful,” “Slightly useful,” “Moderately useful,” “Very useful,” and “Extremely useful.”

*Willingness to share*: Measured with “How likely are you to share this research with others?” The response options were “Not likely at all,” “Slightly likely,” “Moderately likely,” “Very likely,” and “Extremely likely.”

**2.4.1.4 Ethicality/transparency** *Perceived ethicality of researcher*: Measured with “How ethical do you think it is for the authors to use AI in the way they described?” The response options were “Not ethical at all,” “Slightly ethi-

cal,” “Moderately ethical,” “Very ethical,” and “Extremely ethical.”

*Perceived transparency of paper:* Measured with “How transparent do you think this paper is?” The response options are “Not transparent at all,” “Slightly transparent,” “Moderately transparent,” “Very transparent,” and “Extremely transparent.”

## 2.4.2 Moderators

**2.4.2.1 Frequency of using Gen AI** Measured with “Over the last six months, how frequently have you used generative AI?” The response options were “Not at all frequently,” “Slightly frequently,” “Moderately frequently,” “Very frequently,” and “Extremely frequently.”

**2.4.2.2 Knowledge about Gen AI** Measured with “How much do you feel you know about generative AI?” The response options are “None at all,” “A little,” “A moderate amount,” “A lot,” and “A great deal.”

**2.4.2.3 Trust in science** Measured with “To what extent do you trust science overall?” The response options were “None at all,” “A little,” “A moderate amount,” “A lot,” and “A great deal.”

## 2.4.3 Demographics

**2.4.3.1 Gender** Measured with “Which gender do you identify with?” Response options were “Male,” “Female,” “Non-binary/third gender,” and “Prefer not to say.” A variable “female” was created so that respondents who chose “Female” were coded 1 and 0 otherwise.

**2.4.3.2 Age** Measured with “In which year were you born?” A categorical “age” variable was created: [18, 29], (29,39], (39,49], (49,59], (59,69], and above 69.

**2.4.3.3 Education** Participants were asked, “What is the highest level of school you have completed?” Responses were coded as a categorical variable with five categories: “Less than high school,” “High school graduate,”

“Some college/Associate degree,” “College graduate,” and “Post-graduate.”

**2.4.3.4 Income** Participants were asked to identify the total income for themselves and all members of their family who lived with them during the previous year before taxes. Responses were coded as a categorical variable with six categories: “Less than 25 k,” “25 k to 49,999,” “50 k to 74,999,” “75 k to 99,999,” “100 k to 149,999” and “150 k or more.”

**2.4.3.5 Hispanic ethnicity** Respondents were asked, “Are you of Hispanic, Latino, or Spanish origin?” A variable, “Hispanic,” was created and coded 1 if the participant answered “Yes” and 0 if not.

**2.4.3.6 Race** Respondents were asked if they consider themselves to be from the following list: “White,” “Black or African American,” “American Indian or Alaska Native,” “Asian,” and “Native Hawaiian or Pacific Islander.” A dummy variable was created for each response option and coded 1 if the respondent selected that option.

## 2.5 Analysis

$H_1$  to  $H_4$  centered on perceived trust, quality of research, and level of engagement. We had directional hypotheses on eight variables (from Trust in Research to Willingness to Share in the Measurement section), and analyses on perceived ethicality and transparency were exploratory.  $H_1$  to  $H_4$  were tested using ordinary least squares (OLS) regression. The dependent variables included participants’ trust in researchers, trust in findings, perceived skillfulness of the researcher, perceived quality of the research, level of interest, perceived importance, perceived usefulness, and willingness to share the research, with a separate model estimated for each outcome. The key independent variable, AI Statement<sub>*i*</sub>, was a categorical variable indicating the experimental condition to which each respondent was assigned. Control variables ( $X_i$ ) included gender, age, education, income, Hispanic ethnicity, and race.

$$Y_i = \alpha + \tau \text{AI Statement}_i + \beta X_i + \varepsilon_i, \quad (1)$$

For  $H_1$  and  $H_3$ , the reference group was the condition that saw “This research was conducted entirely by the authors without any help from any generative AI tools.” For  $H_2$ , the reference group was the condition that saw “The authors

used generative AI tools to help with methodological tasks such as statistical analyses and coding. All theoretical framing, literature review, and hypotheses were developed by the authors.” For  $H_4$ , the reference group was the condition that saw “The authors used generative AI tools to assist with language polishing.”

$H_5$  was tested with the interaction between the AI statement condition and respondents’ frequency of AI use. We estimated separate models for each dependent variable.

$$Y_i = \alpha + \tau_1 AI\ Statement_i + \tau_2 AI\ Use_i + \tau_3 (AI\ Statement_i \times AI\ Use_i) + \beta X_i + \varepsilon_i, \quad (2)$$

The design, measurement, and analysis plan were pre-registered at OSF (<https://osf.io/grxv3>). The data and code needed to replicate the current research are also available at the same OSF repository.

**Table 1** Comparison of subgroup means across conditions

|                                      | Is no-AI group mean the highest? | No-AI vs. all other groups | Is both-assistance group mean the lowest? | Both-assistance vs. all other groups |
|--------------------------------------|----------------------------------|----------------------------|-------------------------------------------|--------------------------------------|
| <i>Trust</i>                         |                                  |                            |                                           |                                      |
| Trust in researcher                  | Yes                              | 0.17***                    | Yes                                       | -0.11***                             |
| Trust in findings                    | Yes                              | 0.13***                    | Yes                                       | -0.13***                             |
| <i>Competence/quality</i>            |                                  |                            |                                           |                                      |
| Perceived skillfulness of researcher | Yes                              | 0.12***                    | Yes                                       | -0.17***                             |
| Perceived quality of paper           | Yes                              | 0.13***                    | Yes                                       | -0.14***                             |
| <i>Level of engagement</i>           |                                  |                            |                                           |                                      |
| Level of interest in research        | Yes                              | 0.15***                    | Yes                                       | -0.14***                             |
| Perceived importance of research     | Yes                              | 0.12***                    | Yes                                       | -0.11***                             |
| Perceived usefulness of research     | Yes                              | 0.11***                    | Yes                                       | -0.10***                             |
| Willingness to share                 | Yes                              | 0.11***                    | Yes                                       | -0.11***                             |
| <i>Ethicality/transparency</i>       |                                  |                            |                                           |                                      |
| Perceived ethicality of researcher   | Yes                              | 0.24***                    | Yes                                       | -0.20***                             |
| Perceived transparency of paper      | Yes                              | 0.08**                     | No                                        | -0.05*                               |

Columns 2 and 4 report the OLS regression coefficients for dummy variables indicating membership in the No AI group and the Both Assistance group. All models control for respondents’ demographics  
\* $p < 0.05$ ; \*\* $p < 0.01$ ; \*\*\* $p < 0.001$

### 3 Results

#### 3.1 Strong preference for human-only research and penalty for full AI use

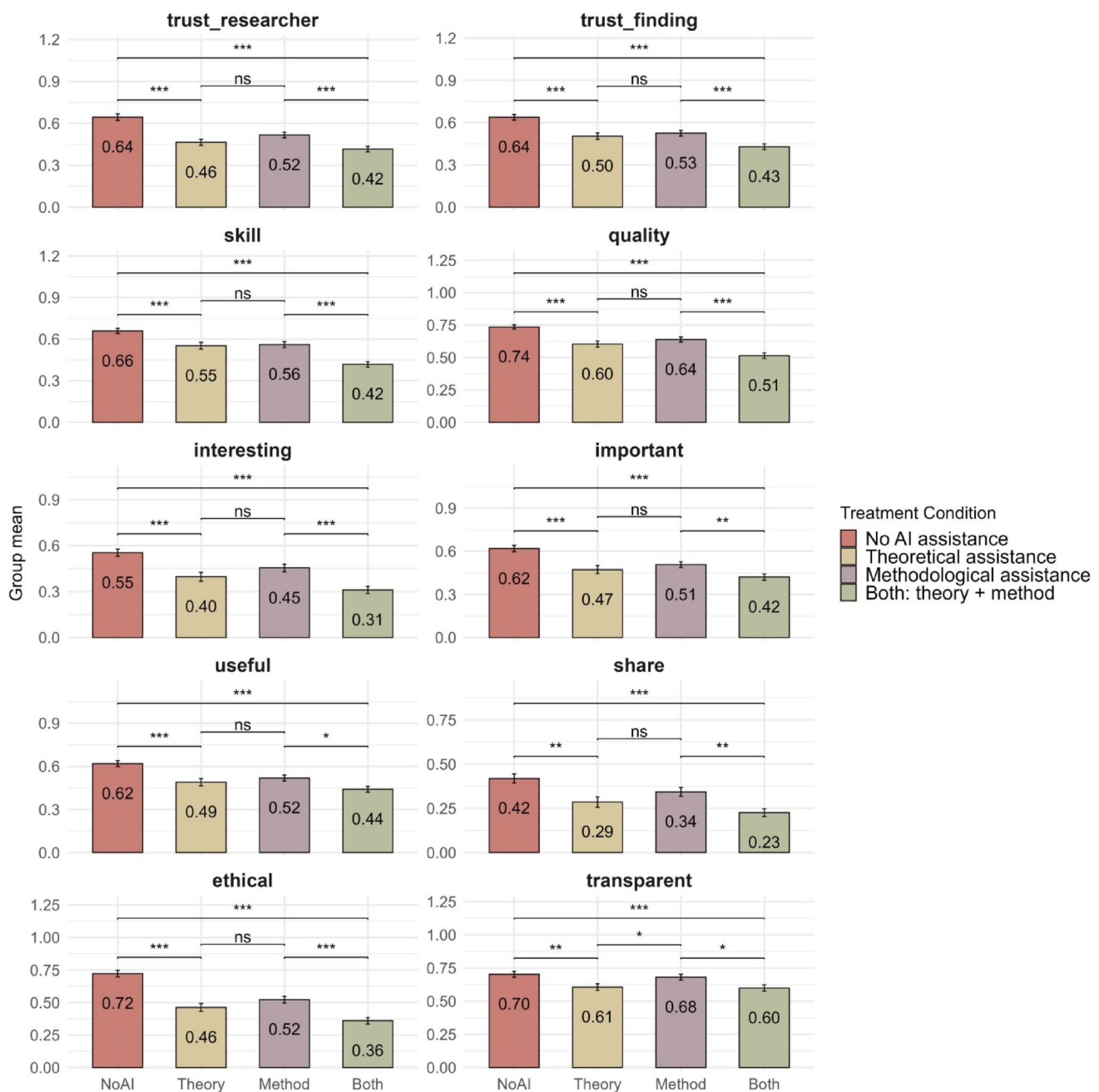
Respondents reported the highest levels of trust, perceived quality, and level of engagement when they learned that the researchers did not use generative AI (Table 1). The “No-AI” group showed the highest mean scores across all ten dependent variables, fully supporting  $H_1$ . Among the eight variables for which directional hypotheses were specified, the “Both Theoretical and Methodological Assistance” group consistently exhibited the lowest mean scores, fully supporting  $H_3$ . Respondents reported the lowest levels of trust, perceived quality, and level of engagement when they learned that the researchers used generative AI to assist both theoretical and methodological tasks. For the two exploratory variables, the perceived ethicality was also lowest for the “Both Theoretical and Methodological Assistance” group. The perceived transparency was lowest for the “Language Polishing” group: Mean=0.58, SD=0.27 (Mean No-AI=0.70, SD=0.25; Mean Both=0.60, SD=0.28).

#### 3.2 Theoretical versus methodological assistance

Compared with “Theoretical Assistance,” although subgroup means (both unconditional and conditional on demographics) were always higher for the “Methodological Assistance” group, none of the eight dependent variables with directional hypotheses showed statistically significant difference (Fig. 1). The effect sizes were small: Cohen’s  $d$  ranged from 0.03 (Perceived Skillfulness of Researcher:  $B=0.01$ ,  $t=0.32$ ,  $p=.75$ ) to 0.22 (Trust in researchers:  $B=0.06$ ,  $t=1.92$ ,  $p=.06$ ), showing limited evidence for  $H_2$ . Perceived ethicality did not differ across the “Theoretical Assistance” and the “Methodological Assistance” group (Mean Theory=0.46, SD=0.33, Mean Method=0.52, SD=0.32,  $B=0.06$ ,  $t=1.63$ ,  $p=0.10$ ), but respondents perceived papers reporting use of generative AI for methodological use as more transparent than for theoretical use (Mean Theory=0.61, SD=0.27, Mean Method=0.68, SD=0.26,  $B=0.07$ ,  $t=2.21$ ,  $p=0.03$ ). Full pairwise contrasts between conditions are reported in the [Supplementary Material](#).

#### 3.3 Reassuring human intellectual contribution

Compared with research using Gen AI for language polishing (“The authors used generative AI tools to assist with language polishing.”), adding “The theoretical framing, hypotheses, analyses, and interpretation were developed entirely by the authors without any help from any generative

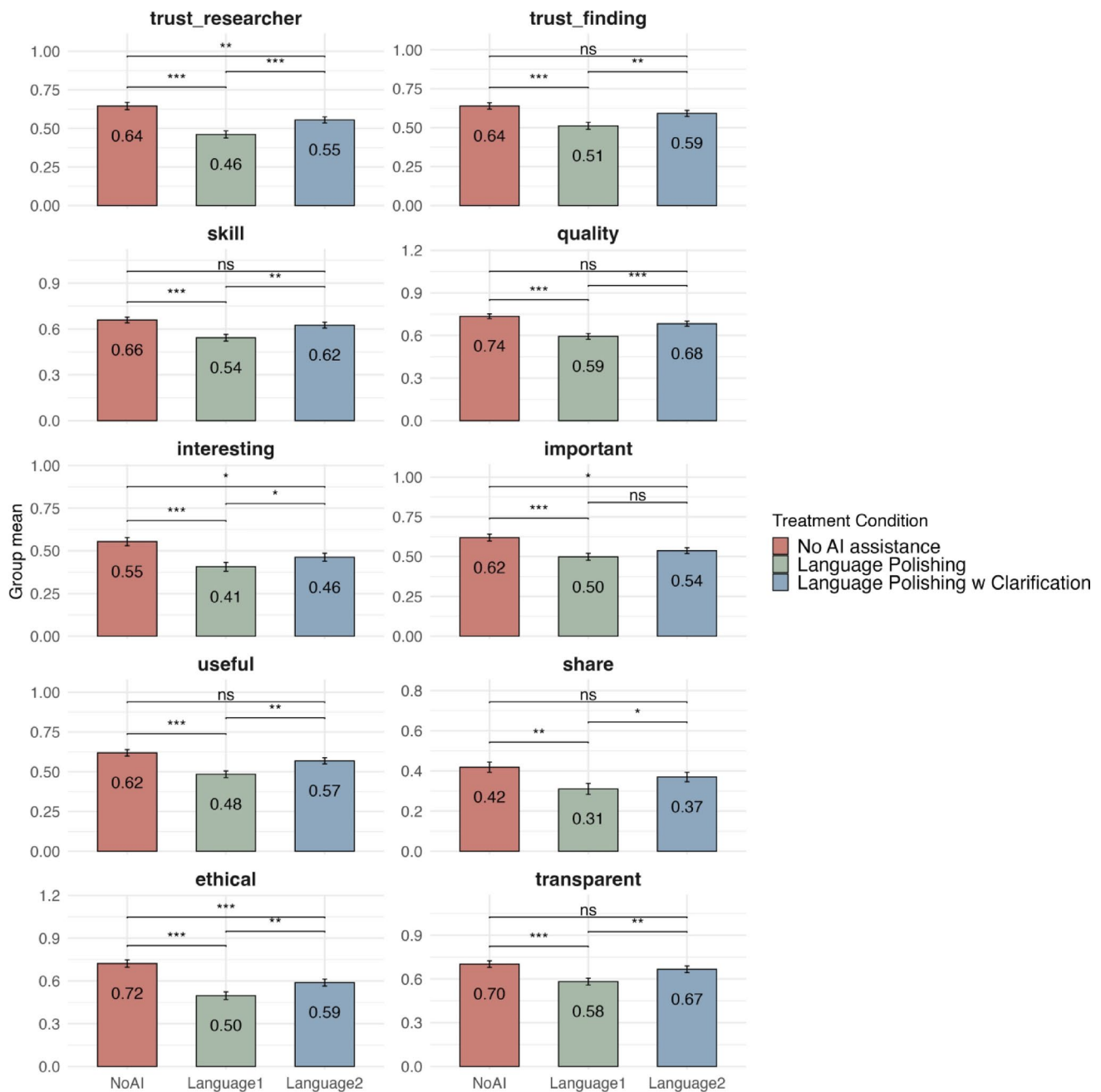


**Fig. 1** Mean ratings across No AI assistance, theoretical assistance, methodological assistance, and both (theoretical+methodological assistance). Bars represent unconditional means with  $\pm 1$  standard error. Significance indicators are based on OLS regressions with demographic controls

AI tools.” significantly increased people’s trust in researchers, trust in findings, perceived skillfulness of researcher, perceived quality of research, level of interest, perceived usefulness, and willingness to share the research. There was no outcome where the clarification condition performs worse than language polishing alone (Fig. 2). H4a was supported, and H4b was not. Respondents also perceived the paper to be more ethical (Mean Language 1=0.50, SD=0.30, Mean Language w Clarification=0.59, SD=0.30,  $B=0.10$ ,  $t=2.71$ ,  $p=0.007$ ) and transparent (Mean Language

1=0.58, SD=0.27, Mean Language w Clarification=0.67, SD=0.27,  $B=0.09$ ,  $t=2.64$ ,  $p=0.009$ ) when the disclosure statement included “The theoretical framing, hypotheses, analyses, and interpretation were developed entirely by the authors without any help from any generative AI tools.”

Respondents penalized researchers even if researchers only used generative AI to assist with language polishing. For all 10 variables, means were significantly higher for the “No-AI” group than for the “Language Polishing” group. Adding “The theoretical framing, hypotheses, analyses, and



**Fig. 2** Mean ratings across No AI assistance, language polishing, and language polishing with additional clarification. Bars represent unconditional means with  $\pm 1$  standard error. Significance indicators are based on OLS regressions with demographic controls

interpretation were developed entirely by the authors without any help from any generative AI tools” closed the gap between No-AI and Language Polishing for six out of ten variables, though respondents still trusted researcher more if they did not use generative AI (Mean No AI=0.65, SD=0.26, Mean Language w Clarification=0.67, SD=0.27,  $B=0.09$ ,  $t=2.93$ ,  $p=0.003$ ), perceived the research to be more interesting (Mean No AI=0.55, SD=0.27, Mean Language w Clarification=0.46, SD=0.29,  $B=0.08$ ,  $t=2.16$ ,  $p=0.03$ ), important (Mean No AI=0.62, SD=0.22, Mean Language w

Clarification=0.54, SD=0.23,  $B=0.06$ ,  $t=2.00$ ,  $p=0.046$ ), and ethical (Mean No AI=0.72, SD=0.29, Mean Language w Clarification=0.59, SD=0.30,  $B=0.13$ ,  $t=3.53$ ,  $p<.001$ ).

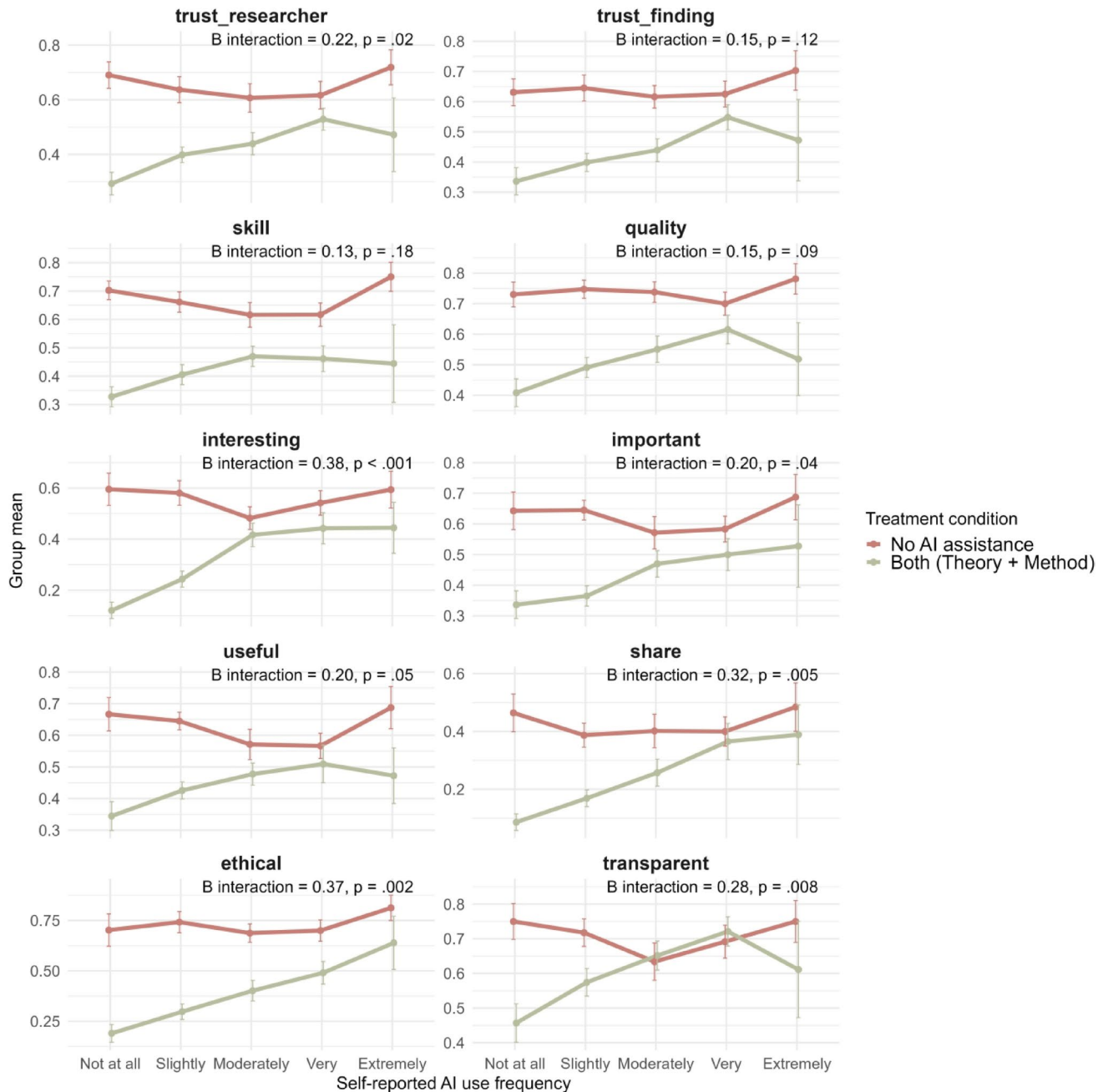
### 3.4 Moderators of AI evaluations

Respondents were more lenient toward researchers’ generative AI use when respondents themselves used generative AI more frequently. The gap between “No-AI” and “Both Theoretical and Methodological Assistance” tended

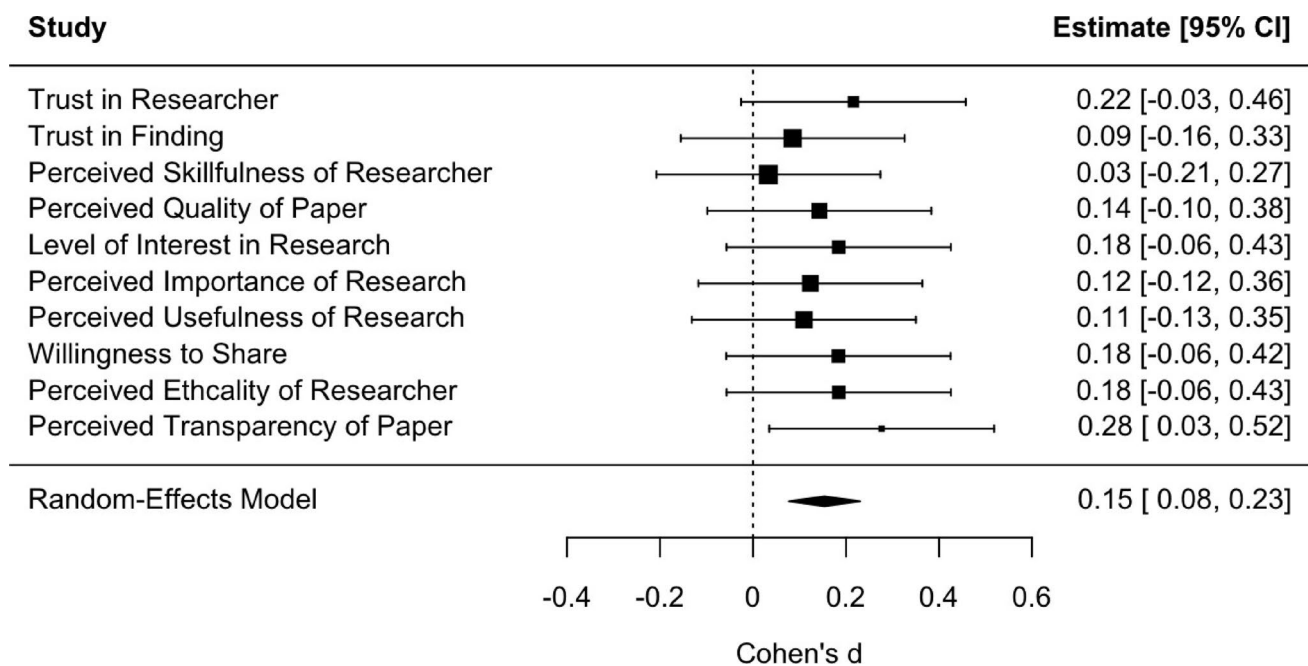
to shrink as respondents' frequency of generative AI use increased (Fig. 3). The interaction terms were positive for all 10 variables, with 6 reaching statistical significance (Trust in researchers, Level of interest in research, Perceived importance, Willingness to share, Perceived ethicality, and Perceived transparency) and two marginally significant (Perceived quality of research and Perceived usefulness of the research).  $H_5$  was partially supported.

Figure 3 seemed to have suggested that greater self-reported AI use was associated with a narrowing of the

evaluation gap up to a point, followed by a potential widening among the most frequent users. However, this pattern was not supported by formal statistical tests. When restricting the analysis to respondents who reported using AI "very" or "extremely" frequently and testing whether the treatment gap differed between these two groups, none of the ten moderation models yielded statistically significant interaction effects. Nonetheless, the number of respondents in the extremely frequent category was very small ( $N=16$  in the No-AI condition and  $N=9$  in the Both-Assistance



**Fig. 3** Subgroup means of outcome variables across AI use frequency and treatment conditions. Points represent unconditional subgroup means, and error bars show  $\pm 1$  standard error.  $p$  values are based on OLS regressions with demographic controls



**Fig. 4** Forest plot of effect sizes (Cohen's *d*) for ten dependent variables

condition), making the results more sensitive to random fluctuation. Accordingly, we refrained from drawing strong conclusions about potential divergence among the most frequent users.

There was little evidence that self-reported AI knowledge was a meaningful moderator. Self-reported AI knowledge did not moderate the treatment effects; none of the ten interaction terms between treatment (“No-AI” v. “Both Theoretical and Methodological Assistance”) and AI knowledge were statistically significant. Trust in science was not a meaningful moderator, either. None of the ten treatment  $\times$  trust in science interactions were statistically significant. Detailed test statistics are available in the [Supplementary Material](#).

### 3.5 Non-preregistered meta analysis

Because the means (both unconditional and conditional on demographics) for all eight directional outcomes were consistently higher in the “Methodological Assistance” condition than in the “Theoretical Assistance” condition, we conducted an exploratory, non-preregistered meta-analysis to assess whether this small but systematic advantage for methodological use might nonetheless be detectable when aggregating across outcomes. A post-hoc sensitivity analysis indicated that the available sample sizes ( $N=120$  for “Theoretical Assistance”;  $N=148$  for “Methodological Assistance”) provided 80% power to detect an effect of Cohen's  $d=0.35$ , a value between a small and medium effect [10].

However, the observed effect sizes for all outcomes were modest—none exceeding 0.22. Thus, the absence of statistical significance may not be taken as evidence of equivalence but a reflection of the limited power to detect effects of this small magnitude.

The summary effect sizes were small but significantly different from zero:  $d=0.13$  for the eight variables with directional hypotheses and  $d=0.15$  when all ten variables were included (Fig. 4). In the aggregate, the pattern of effects indicated a small yet non-zero advantage for methodological over theoretical use of generative AI. Respondents were somewhat more accepting of AI when it assisted with computational or procedural tasks rather than with conceptual or interpretive elements of the research process.

## 4 Discussion

The present research is a pre-registered survey experiment ( $N=806$ ) on public perceptions of academic researchers' use of generative AI. It offers a systematic experimental test of the impact of AI assistance on a broad array of credibility and value judgments. Our 10 dependent variables covered trust, competence, level of engagement, and perceived research ethics. These dependent variables examine both perceptions of humans who use AI and the research, the product of AI assistance. We found that the public reported strong preference for human-only research and researcher.

The most negative perceptions were reported for full AI involvement (both for theory assistance and method assistance). Respondents were more lenient toward the use of generative AI for methodological versus theoretical tasks, though the effect sizes were small. Frequency of generative AI use was a strong moderator, while self-reported AI knowledge and trust in science were not.

One lingering concern is whether the dependent variables really captured distinct dimensions of respondents' evaluations of the research. One possibility is that these items are all just variations of "do you think this is good research." An Exploratory Factor Analysis indicated a two-factor structure, with the two Trust variables, two Competence/Quality variables, and perceived ethicality loading on the first factor, and the four variables we conceptualized under the umbrella of "Level of Engagement" loaded onto the second factor (Detailed loading statistics are available in the [Supplementary Material](#)). Perceived transparency was adjacent to the first factor (loading=0.59; comparably weaker than other variables in the first factor), consistent with its distinct theoretical role in evaluations of AI disclosure.

#### 4.1 AI aversion

AI aversion has been documented across many domains [8, 33, 34, 44], and the present study demonstrates that academic research is no exception. Respondents expressed the highest levels of trust, perceived quality, and level of engagement when they learned that the research was conducted entirely without generative AI.

One plausible explanation for these negative perceptions is that the perceived value of academic research rests on its association with human creativity, intellectual labor, and conceptual innovation [27]. The involvement of generative AI, particularly in tasks central to reasoning and interpretation, may be viewed as diminishing these human contributions and obscuring individual accountability [44]. Respondents may have also questioned the competence of AI, despite much literature suggesting that AI aversion persists even when AI performs better than humans [13, 33, 44, 64].

The negative perceptions extended beyond the research itself (the product of AI use) to encompass the researchers who employed such tools. Respondents interpreted the use of generative AI as a reflection of the researcher's character. Our results resonate with recent research on how AI use hurts social evaluations of humans who use AI. People who got help from AI were perceived to be lazier, less competent, and less diligent than people who got help from human sources [47]. People who got help from AI were even "dehumanized" in the sense that they were ascribed less emotionality and rational agency [11].

#### 4.2 Methodological and theoretical use

Ratings of research and researchers declined as the technology involved more with core intellectual tasks. Using AI for methodological assistance was viewed as somewhat more acceptable than using it for theoretical assistance, likely because statistical coding and computation are already seen as domains of automation and technical support rather than intellectual ownership [15]. In contrast, theoretical tasks lie more at the heart of scholarly creativity and judgment [27]. Delegating these activities to AI may therefore violate implicit moral norms about what constitutes legitimate authorship and responsible scientific practice.

These patterns can be further interpreted through the lens of attribution theory. Attribution theory posits that evaluators assess outcomes based on whether they attribute them to internal causes, such as effort, skill, and intelligence, or to external causes, such as tools or assistance [23]. In academic research, theoretical activities are widely perceived as the core locus of scientific creativity and intellectual contribution [27, 43]. When AI is disclosed as assisting with these tasks, audiences may shift causal attribution away from the researcher and toward the technology, thereby reducing perceived competence, trust, and legitimacy.

#### 4.3 AI for language polishing

Although some scholars have expressed optimism on how AI language tools may close the gap between native and non-native speakers [37], the current research found that respondents penalized research and researchers who used AI for mere language polishing. Linguistic assistance was not interpreted as a neutral accessibility aid but as a signal of diminished competence, resonating with recent research on how norms of academic legitimacy remained closely tied to native-like English fluency [28, 42]. AI is not leveling the play-field because disclosure of language-polishing AI exposes non-native scholars to new forms of scrutiny.

Explicitly stating that core intellectual tasks were conducted entirely by human authors partially (but not fully) closed the trust gap. On one hand, these results are somewhat reassuring because they suggest the public's concerns were not rooted in general technophobia, but rather in perceptions of what constitutes legitimate academic labor. Clear disclosure and boundary setting restored some faith in research and researcher. However, the clarification was still only partially effective, meaning the public held a somewhat exclusionary standard of academic legitimacy. Once AI use is disclosed, some uncertainty lingers, and researchers are never fully restored to equal standing with those who can credibly claim unassisted fluency.

Among the six conditions, “AI for Language Polishing” was rated as the least transparent. Two possible reasons might account for this. First, respondents might be dissatisfied with the scope of AI use disclosed. They may desire more details on what kind of language polishing (e.g., fixing grammar, rephrasing, reframing) was used. This aligns with our earlier logic of how audiences scrutinize researchers through a legitimacy lens. AI assistance is perceived to be normatively risky, so respondents demand more precision and justification. Second, respondents may have inferred that the statement concealed broader or undisclosed uses of AI. Readers might suspect that the true AI use was more than what was disclosed. This possibility is consistent with prior work on disclosure and trust, which suggests that when transparency cues are incomplete or ambiguous, audiences often interpret them as signs of strategic omission or dishonesty rather than honesty [46, 55].

The cynical hypothesis that additional clarification of AI use might backfire by heightening suspicion was not supported. This suggests that the public’s negative reactions to AI use are not driven primarily by a generalized tendency to punish transparency or by reflexive suspicion of defensive disclosure. Instead, the results indicate that audiences are capable of making meaningful distinctions about the scope and location of AI involvement. When clarification explicitly reaffirmed that core intellectual tasks remained human-driven, respondents can evaluate in a more favorable direction rather than interpreting the clarification as an attempt at obfuscation.

#### 4.4 Frequency of AI use as a moderator

Respondents who reported using AI more frequently were more lenient toward AI-assisted researchers. Frequent users likely possess greater awareness of AI’s capabilities and limitations [25], and thus perceive its use as pragmatic rather than ethically questionable. This resonates with previous literature indicating frequent AI users are more likely to give positive ratings to AI-assisted products or humans that seek assistance from AI. In somewhat similar veins, Chen and Cui [8] found that frequent AI users were more lenient toward doctors who used AI to assist their diagnosis. Reif et al. [47] found that frequent AI users were less punitive toward AI-using candidates: the penalty shrinks as the evaluator’s own AI use increased.

The present study found little evidence for a moderating effect of self-reported AI knowledge, consistent with Chen and Cui [8], who likewise observed that knowledge failed to close the trust gap between physicians who used AI to assist diagnosis and those who did not. Knowledge of AI (e.g., how AI functions) may not be sufficient to alter people’s moral or affective responses to its use in medical

domains or science. Perceptions of the use of generative AI in health and science arise less from technical literacy than from intuitive and value-laden judgments about authenticity, fairness, and human agency [33]. Thus, knowing how AI operates may help individuals recognize its capabilities but does little to mitigate underlying concerns that it displaces human creativity or accountability. Public aversion to AI is rooted more in perceived normative boundaries, i.e., what kinds of tasks are appropriate for human versus machine agents, than in informational deficits.

The divergence between knowledge and usage as moderators suggests that experience rather than cognitive understanding shapes attitudes toward AI-assisted research. Self-reported knowledge captures a declarative awareness of how AI works, but such abstract understanding did not translate into emotional comfort or moral acceptance. In contrast, frequent users interact with AI systems directly, allowing them to observe both their limitations and practical utility first-hand. Experiences help normalize AI as a functional tool rather than a symbolic threat to human creativity or integrity. Prior research on technology adoption similarly shows that hands-on engagement, rather than technical literacy, reduces perceived risk and strengthens acceptance [12, 49]. Hence, respondents’ personal experience with AI use may have cultivated trust through habituation, while knowledge alone lacked the affective and behavioral grounding necessary to influence moral evaluations.

Another possible explanation for the moderating role of frequency of use and the lack of moderation by self-reported knowledge is cognitive dissonance. Individuals are motivated to resolve psychological discomfort caused by inconsistency between their attitudes and behaviors by adjusting one to align with the other [19]. In our context, frequent AI users may experience cognitive dissonance if they hold negative views about AI-assisted work while frequently engaging in such practices themselves. To reduce this discomfort, they may adjust their attitudes to be more favorable toward AI use for both themselves and others. As a result, frequent users may adopt more forgiving evaluations because of a desire to avoid negative self-implications. This resonates with recent empirical studies across different technology contexts suggesting cognitive dissonance can motivate adjustments in both attitude and behavior to maintain psychological consistency [36, 48].

As an exploratory analysis, we found a limited moderating role of general trust in science. Parallel to the limited moderating role of self-reported AI knowledge, confidence in the scientific enterprise does not necessarily translate into acceptance of AI-assisted research. Together, these findings indicate that direct engagement or identity-consistent judgment, rather than generalized trust or understanding, play a central role in shaping responses to AI disclosure.

## 4.5 Practical implications

The present research points to a potential erosion of public trust and engagement in science as generative AI becomes more visibly integrated into the research process. Even limited AI involvement like language polishing can reduce trust, level of engagement, and perceived re-search ethics. Our findings illustrated a transparency dilemma: disclosure of AI use is motivated by ethical norms of openness and accountability, yet such transparency reduces trust.

Generative AI will become more integrated into human lives, and academic research will be no exception. The public's willingness to engage with scientific findings may be further compromised by the integration of generative AI. The challenge for the scientific community will be to preserve the perception of human intellect and integrity that underlies its social legitimacy.

Clear boundary-setting around human intellectual labor can mitigate but not eliminate penalties. Explicitly stating that intellectual tasks were conducted entirely by human authors consistently improved evaluations relative to language-polishing disclosure alone. However, because clarification only partially closed the gap, such boundary-setting should be viewed as a harm-reduction strategy rather than a complete solution.

If the boundary between human and machine authorship continues to blur, or if reassurance remains incomplete, the public may increasingly question whether science still reflects human curiosity, judgment, and moral accountability. Such doubts risk deepening the existing crisis of confidence in scientific institutions [21], transforming skepticism about individual studies into broader disengagement from the scientific enterprise itself.

The present research also found that those with greater personal experience using AI are more accepting, while general scientific trust or abstract knowledge about AI provide little buffer. Education effort on how AI works (see [7] for a systematic review on AI literacy) may be insufficient to preserve trust in science. Future science communication may need to move from “explaining AI” to “letting people experience AI” to preserve trust in and engagement with science.

## 4.6 Limitation and future research

A potential limitation is that participants did not evaluate an actual research article but instead made judgments based on a brief hypothetical scenario with an AI disclosure statement. As a result, our findings reflect expectations about a paper and its authors rather than evaluations based on substantive content. The current research design helped isolate the effect of AI disclosure itself, independent of variation

in article topic, quality, or writing style. Presenting a full article would have introduced substantial confounds and would have made it difficult to attribute observed effects specifically to the disclosure statement. Additionally, prior research suggests that non-expert audiences typically lack the technical background needed to independently assess methodological rigor or theoretical contribution and therefore rely heavily on contextual cues such as authorship, institutional affiliation, and disclosure practices [39, 60]. In this respect, the current design reflects how most members of the public encounter academic research in practice. Future research could examine how AI disclosures interact with substantive article quality and characteristics. Nevertheless, incorporating real articles poses challenges, as lay audiences may still rely on heuristics, and any specific article may introduce idiosyncratic features that limit external and ecological validity.

Future research may also examine whether the moderating role of AI experience is non-linear. In the present research, we did not draw strong conclusions about potential divergence among the most frequent users because the small size of the extreme-use subgroup renders its estimates more sensitive to random fluctuation and outliers. Future research can zoom in onto the potential non-linear effect to assess whether extremely frequently AI use is systematically associated with heightened critical scrutiny and boundary drawing.

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**Data availability** Data and code needed to replicate the current research are available at <https://osf.io/grxv3>.

## Declarations

**Competing interests** The authors declare no competing interests.

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